# MILITARY SPECIFICATION

## RECORDER-REPRODUCER SET, SOUND AN/AQH-1(V)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

## 1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide an airborne magnetic tape recording/reproducing instrumentation system suitable for ASW data recording, storage and reproduction. The equipment specified herein shall be capable of accomplishing this function through the use of direct recording and wide band FM recording techniques as specified in this document.

1.2 Classification - The Recorder-Reproducer Set, Sound, AN/AQH-1(V) shall consist of the following items:

| Item | Type Designation | Appl. Para. |
|---|---|---|
| Tape Transport - Power Supply | MX-4826/AQH-1(V) | 3.4.1 |
| Chassis, Amplifier | CH-506/AQH-1(V) | 3.4.2 |
| Amplifier, Recorder-Reproducer, Direct | AM-3769/AQH-1(V) | 3.4.3 |
| Amplifier, Recorder-Reproducer, FM | AM-3770/AQH-1(V) | 3.4.4 |
| Amplifier, Recorder-Reproducer, ICS | AM-3773/AQH-1(V) | 3.4.5 |
| Mounting (Transport) | MT-3025/AQH-1(V) | 3.4.6 |
| Mounting (Amplifier Chassis, dual) | MT-3024/AQH-1(V) | 3.4.7 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

1.3 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment: AN/ARR-52, AN/AQA-5, AN/AQA-1, AN/ASA-20, AN/AIC-22(V).

## 2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-C-172 | Cases; Bases, Mounting; and Mounts, Vibration (For use with Electronic Equipment in Aircraft) |
| MIL-E-4682 | Electron Tubes and Transistors, Choice and Application of |
| MIL-W-5088 | Wiring; Aircraft, Installation of |
| MIL-E-5400 | Electronic Equipment, Aircraft, General Specification for |
| MIL-T-5422 | Testing, Environmental, Aircraft Electronic Equipment |
| MIL-I-6181 | Interference Control Requirements, Aircraft Equipment |
| MIL-M-7793 | Meter, Time Totalizing |
| MIL-E-17555 | Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of |
| MIL-T-18303 | Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for |
| MIL-N-18307 | Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

MIL-T-21029 Tapes, Instrumentation, Recording Magnetic Oxide

STANDARDS

Military

MIL-STD-704 Electric Power, Aircraft Characteristics and Utilization of

MS-17322 Meter, Time Totalizing, Miniature, Digital 115 Volt 400 cycle

IRIG

IRIG Document 106-60 Telemetry Standards

PUBLICATIONS

Bureau of Naval Weapons

EI -440 Avionics Installation Instructions for Sound Recorder - Reproducer Set AN/AQH-1(V)

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code DCI, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.



Source: https://assist.dla.mil -- Downloaded: 202610-26T04:33Z
Check the source to verify that this is the current version before use.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to failure of more than 5,000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable subassemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.2.1 Equipment Description - The equipment required for each Recorder-Reproducer System shall be comprised of the following. Two System Configurations shall be possible.

| Items | System I 14 Channel maximum | | System II 7 Channel maximum | |
|---|---|---|---|---|
| | Quan. | Type | Quan. | Type |
| Transport - Power Supply | 1 | MX-4826 | 1 | MX-4826 |
| Chassis, Amplifier | 2 | CH-506 | 1 | CH-506 |
| Amplifier, Recorder-Reproducer Direct | * | AM-3769 | * | AM-3769 |
| Amplifier, Recorder-Reproducer FM | * | AM-3770 | * | AM-3770 |
| Amplifier, Recorder-Reproducer ICS | * | AM-3773 | * | AM-3773 |
| Mounting (Transport) | 1 | MT-3025 | 1 | MT-3025 |
| Mounting (Amplifier Chassis) | 1 | MT-3024 | 1 | (not assigned) |

*Direct, FM, and ICS Amplifiers may be used interchangeably and in any combination up to the maximum total for the system.



Source: https://assist.dla.mil -- Downloaded: 202640-26T04:33Z
Check the source to verify that this is the current version before use.

3.2.2 Amplifiers - The equipment shall utilize Direct and Frequency Modulation (FM) type recording and reproducing techniques. All record and reproduce amplifiers shall be of the current state-of-the-art design consistent with maintaining a high input impedance. The choice of recording/reproducing Direct, FM or ICS shall be determined by plugging the appropriate amplifier in any one of the record/reproduce amplifier positions of the Amplifier Chassis.

3.2.3 Construction - The equipment shall be constructed in keeping with a modular design concept. Plug-in units and plug-in assemblies shall be used wherever possible. The recording/reproducing requirements shall be met through the use of individual plug-in magnetic head assemblies whose precision and construction will allow their interchange or replacement by inexperienced personnel without the need for shims, adjustments, soldering or alignment of any kind.

3.2.4 Total Weight - The total weight of the equipment shall be a minimum consistent with good design. The weights of the individual items of 1.2 shall not exceed the weights as specified in the applicable paragraphs under detail requirements.

3.2.5 Reliability

3.2.5.1 Operationable Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 200 hours without the necessity for readjustment of any controls or assembly other than the supply of tape. Intermittent operation may be accompanied by allowable warm up time as specified in paragraph 3.2.13 of this equipment specification. Normal tape path cleaning is not to be interpreted as a readjustment.

3.2.5.2 Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment, and the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.2.5.3 Reliability in Mean Time Between Failures (MTBF) - The equipment shall have a minimum of 300 hours of mean time between failures. A failure shall be defined as any malfunction which causes the equipment to fail to meet the minimum performance requirements specified for the equipment.



Source: https://assist.dla.mil -- Downloaded: 202640-26T04:33Z
Check the source to verify that this is the current version before use.

3.2.5.4 Time Totalizing Meters - The equipment shall contain time totalizing meters in accordance with Specification MIL-M-7793. Two meters shall be included in the following unit, one of which shall monitor total system operating time and the other shall monitor total tape travel time. Each meter shall be clearly identified regarding its respective function.

| Unit | Type of Meter |
|---|---|
| Transport-Power Supply MX-4826 | MS 17322 (2 each) |

3.2.6 Cabling and Connections -

3.2.6.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.6.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment. In any event the cable lengths between Amplifier Chassis CH-506 and the Tape Transport - Power Supply MX 4826 shall not exceed ten feet.

3.2.7 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.8 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.9 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.



Source: https://assist.dla.mil -- Downloaded: 202610-26T04:33Z
Check the source to verify that this is the currentversion before use.

3.2.10 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.11 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

| | |
|---|---|
| Temperature | Room ambient (25°C ± 5°C) |
| Altitude | Normal ground |
| Vibration | None |
| Humidity | Room ambient up to 90% relative humidity |
| Input power voltage | 115 VAC ± 1%, 400 CPS ± 1%<br>3 phase grounded neutral |

3.2.12 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment, except as modified herein.

3.2.12.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400, Figure 5, except as follows:

(1) Vibration with Isolators - Compliance to specification for FM Signal-to-Noise under the vibration requirements of Curve I may be limited to a maximum of 5 g input force. The equipment shall be operated to the limit of 10 g input force but need not meet performance above 5 g's; however, on completion of the test the system shall demonstrate performance.

(2) Vibration with Isolators Removed - The equipment shall be operated during vibration per Curve II but need not meet performance. However, on completion of the test the system shall demonstrate performance.

3.2.12.2 Temperature - A minimum temperature of -30°C in lieu of -54°C is permitted for compliance with Direct Signal-to-Noise requirements of paragraph 3.3.5.4. A linear degradation to 36 db is allowed from -30°C to -54°C.



Source: https://assist.dla.mil -- Downloaded: 202610-26T04:33Z
Check the source to verify that this is the current version before use.

3.2.13 Warm-Up Time - The time required for the equipment to warm up prior to operation within specification shall not exceed 5 minutes under standard conditions and 15 minutes at extreme service conditions, except that warm-up time for tape supply between -20°C and -54°C shall not exceed 30 minutes.

3.2.14 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD 704 except as modified herein and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power shall not exceed the specified amounts.

(1) AC Power (3 Phase) 115/200V, Category "B" 600VA

3.2.14.1 Power Transients - MIL-STD-704 limit 3 of figure 2 shall not be less than 75 Vrms.

3.2.14.2 AC Overvoltage Protection - The equipment shall not be damaged by overvoltage specified in Figure 2, Limit 1 of MIL-STD-704 and shall automatically resume operation when the voltage returns within limits.

3.2.14.3 AC Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified by MIL-STD-704 and shall automatically resume normal operation when the voltage returns within limits. The equipment shall be capable of operating at 360 CPS for 10 seconds without damage.

3.2.14.4 Power Interruption - The equipment shall withstand a power interruption of 100 milliseconds and shall resume operation without the need of a "reset" after the power has returned to normal conditions.

3.2.15 Standby Provisions - The equipment shall provide for a standby position whereby rated voltage shall be applied to all tube filaments. Except for the associated filament circuits, voltages shall not be applied to other parts of the equipment. If electron tubes are not used, the standby position need not be provided.

3.2.16 External Finish - The equipment shall be finished color number 36231, in accordance with Standard FED-STD-595.

3.3 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions, when using magnetic tape 3M Type 959 or equivalent. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.3.1 Operation - The equipment shall be capable of recording on all channels simultaneously. Concurrently, and without affecting the "record" mode, all channels can be monitored (reproduced) as desired. The equipment shall be capable of reproducing pre-recorded information in a "playback" mode without the tape being acted upon in any way by the record system.

3.3.2 Tape Speeds - Tape speeds of 30, 15, 7 1/2 and 3 3/4 inches per second shall be provided. It shall be possible to electrically select at least one of two tape speeds, either 30 or 15, or 7 1/2 or 3 3/4, both locally on the Tape Transport MX-4826 and at the remote control described in paragraph 3.4.1.5.2. Transition from speeds of 30 and 15 ips to 7 1/2 and 3 3/4 ips may be accomplished by belt changes or other mechanical means at the Tape Transport/Power Supply. If belts are used, provisions shall be made to store spare belts within the Tape Transport/Power Supply. In any event, transition from one to any other tape speed shall not reduce the performance requirements set forth in this equipment specification.

3.3.3 Start Time - With equipment in standby and warm up conditions met, and with the actuation of the "record" function, stable tape operation shall be realized and the equipment shall perform within the limits of normal operating specifications as follows:

| Tape Speed | Start Time (maximum) |
|---|---|
| 3 3/4 ips | 3.0 seconds |
| 7 1/2 ips | 3.0 seconds |
| 15 ips | 5.0 seconds |
| 30 ips | 10.0 seconds |

3.3.4 Frequency Reference Signal - A frequency reference signal shall be provided for recording on the tape. The frequency shall be 10,000 cps ± .1%. The frequency reference signal shall share the same channel as the ICS commentary and shall be directly coupled to the channel in the record mode. The level of the reference signal shall be .5 ± .1 Vrms.

3.3.5 Direct Record/Reproduce Performance

3.3.5.1 Frequency Response - The frequency response shall extend over the bandwidth as indicated for the respective tape speeds and shall be uniform within ± 3 db between band edges.

| Tape Speed | Band Width |
|---|---|
| 3 3/4 ips | 50 to 12,500 CPS |
| 7 1/2 ips | 50 to 25,000 CPS |
| 15 ips | 50 to 50,000 CPS |
| 30 ips | 50 to 100,000 CPS |



Source: https://assist.dla.mil -- Downloaded: 202610-26T04:33Z
Check the source to verify that this is the current version before use.

3.3.5.2 Differential Time Delay - The differential time delay of the output signals between any two direct channels shall not exceed 0.5 milliseconds when tested with a common input to the two channels, in the frequency range of 200 to 5000 CPS.

3.3.5.3 Harmonic Distortion - The harmonic distortion shall be no more than one per cent third harmonic for a normal input record level of 1.5 volts rms, and not more than three per cent third harmonic distortion for a maximum input record level of 3 volts rms. The test frequency shall be 1,000 CPS.

3.3.5.4 Signal to Noise Ratio - The signal to system noise of combined record and playback shall be not less than the values as stated below over the input signal level range as indicated with noise limited to a band pass of 200 to 5000 CPS.

| Input Signal Level | S/N Ratio |
|---|---|
| .15 V rms | 20 db |
| .30 V rms | 26 db |
| .60 V rms | 32 db |
| 1.50 V rms | 40 db |
| 3.00 V rms | 46 db |

3.3.6 FM Record/Reproduce Performance

3.3.6.1 Frequency Response - The frequency response shall extend over the band width as indicated for the respective tape speeds and shall be uniform within ± 0.5 db over the frequency range of 10 to 1,000 CPS and ± 3.0 db from 10 CPS to the high frequency shown for each tape speed.

| Tape Speed | Band Width |
|---|---|
| 3 3/4 ips | 10 to 1,250 cps |
| 7 1/2 ips | 10 to 2,500 cps |
| 15 ips | 10 to 5,000 cps |
| 30 ips | 10 to 10,000 cps |

3.3.6.2 Differential Time Delay - The differential time delay of the output signals between any two FM channels shall not exceed 0.5 milliseconds when tested with a common input to the two channels in the frequency range of 10 to 1000 cps. The differential time delay from 10 to 17 cps shall not result in a relative phase shift greater than 3.0 degrees between any two channels.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.3.6.3 Phase Shift Linearity - Deviation from a linear slope shall be 3.0 degrees maximum over the recording frequency band of 10 to 1,000 cps.

3.3.6.4 Harmonic Distortion - The total harmonic distortion shall be no more than two per cent at frequencies 10 to 1,000 cps for all tape speeds.

3.3.6.5 Signal to Noise Ratio - The signal to system noise per one cycle band width of combined record and playback shall be 65 db or more at frequencies of 10 to 1000 cps at all tape speeds for an output signal level of 1.0 volt rms and shall decrease proportionately, but shall not be less than 49 db for an output signal level of 0.16 volts rms.

3.3.7 ICS Record/Reproduce Performance

3.3.7.1 Frequency Response - The ICS frequency response shall be within ± 3 db from 300 to 3,000 CPS at all tape speeds specified in this equipment specification.

3.3.7.2 Harmonic Distortion - The total harmonic distortion at the maximum input level of 12 Volts rms shall not exceed three per cent.

3.3.7.3 Signal to Noise Ratio - The signal to system noise of combined record and playback shall be not less than 40 db at a reference record output level of 9 volts rms.

3.4 Detail Requirements

3.4.1 Tape Transport - Power Supply MX-4826/AQH-1(V) - The tape transport-power supply unit shall meet the following requirements:

3.4.1.1 Function - The tape transport-power supply unit shall contain the reels, capstans, head assemblies, switching logic, and other mechanical means necessary for handling the magnetic tape. The tape transport-power supply unit shall also convert primary aircraft power to the voltages required for operation of the recorder-reproducer system.

3.4.1.2 Form Factor - The tape transport-power supply unit shall not exceed the following dimensions:

| Width | Height | Length or Depth |
|---|---|---|
| 16.5 inches | 14 inches | 20 inches |



Source: https://assist.dla.mil -- Downloaded: 202610-26T04:33Z
Check the source to verify that this is the currentversion before use.

3.4.1.3 Weight - The weight of the tape transport-power supply shall not exceed 105 pounds.

3.4.1.4 Contents - The tape transport-power supply shall contain the following major assemblies:

(1) Transport Plate Assembly
(2) Power Supply Assembly
(3) Static Inverter Assembly
(4) Control Chassis Assembly

3.4.1.4.1 Transport Plate Assembly - The transport plate assembly shall contain the precision plate subassembly, the capstan motor, the reeling motors, the bias adder network subassembly, local control switches, tape guide mechanism, and elapsed time meters. The following requirements shall be met:

(1) Tape Width - The tape transport shall accommodate one inch wide 1.0 to 1.5 mil mylar magnetic tape.

(2) Reels - The tape transport shall accommodate 10.5 inch precision and/or standard NARTB reels interchangeably.

(3) Rewind - Facilities for reversing tape travel and rewinding the tape onto the supply reel shall be provided. It shall not require more than 3 minutes to rewind 2500 ft. of tape.

(4) Fast Forward - Facilities for fast forward movement of the tape from supply reel to take up reel shall be provided at a rate of speed equal to the rewind speed.

(5) The precision plate subassembly shall include the capstan drive, pinch rollers, magnetic record/reproduce heads, and reproduce head preamplifiers.

3.4.1.4.2 Power Supply Assembly - The power supply shall provide the following from 3 phase aircraft power:

(1) Regulated +20 VDC and -20 VDC.
(2) Unregulated +28 VDC for relay operation
(3) +115 VDC for reel motor operation

3.4.1.4.3 Static Inverter Assembly - The inverter assembly shall provide 115 volt 400 cycle square wave power to the capstan motor.



Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.4.1.4.4 Control Chassis Assembly - The control chassis assembly shall provide switching logic for controlling system operation. The 10 KC reference frequency oscillator shall be located in the control chassis assembly.

3.4.1.4.5 Record and Reproduce Heads - Magnetic heads utilized on the transport described herein shall meet or exceed the requirements of the IRIG Document 106. The following requirements shall also be met:

(1) Head Stack Configuration - The magnetic head stack shall be of the rigid block type. All head assemblies shall be of such precision and design that they can be changed in the field without the need for shimming, adjusting, or alignment of any kind. With such interchange, there shall be no degradation of any of the requirements of this equipment specification. The metal front surface of the head stacks shall present a hard, smooth surface across the entire width of the tape in order to assure a good head to tape contact.

(2) Head Geometry - There shall be a total of four head stacks, two for recording and two for reproduction. Recording and reproducing head stacks shall be so mounted as to minimize electrical coupling.

3.4.1.4.6 Reel Cover - A cover shall be provided to enclose the reels. The tape supply shall be visible without opening or removing the cover from the transport.

3.4.1.5 Controls - The following controls shall be provided as indicated.

3.4.1.5.1 Front Panel Controls - The following push button type controls, labeled as indicated shall be available on the front panel for operational use. The control signal in each case shall be a ground potentia

(1) POWER/READY - This switch shall provide for the local control of power to the equipment.

(2) LOCAL/REMOTE - This switch shall provide for transfer of control of equipment operation to a remote control panel.

(3) SLOW/FAST - This switch shall provide electrical switching between 3 3/4 ips and 7 1/2 ips or 15 ips and 30 ips.



Source: https://assist.dla.mil -- Downloaded: 202640-26T04:33Z
Check the source to verify that this is the current version before use.

(4) RECORD - This switch shall place all channels in the record mode.

(5) REPLAY - This switch shall place all channels in the reproduce or playback mode.

(6) REWIND - This switch shall reverse tape travel. Tape is shuttled from take-up reel to input reel rapidly.

(7) FORWARD - This switch shall place the equipment in a fast forward mode and tape shall be shuttled at fast speed from input reel to take up reel.

(8) STOP - This switch shall stop tape travel and place the equipment in "READY" mode.

3.4.1.5.2 Remote Controls - Capability to control and display all functions remotely except "Rewind" and "Fast Forward" shall be provided. The "Stop" function in remote shall be accomplished by the removal of both the "Record" and "reproduce" electrical grounds. All other control wires shall be a ground potential. Control and display of all remote functions will not be the responsibility of the equipment contractor and is not a part of this specification. The record and reproduce controls shall be independent of each other with no conflict for simultaneous operation of both. When the equipment is in the "Remote" function, all "Local" controls shall be inactive except the "Local/Remote" switch.

3.4.1.5.3 Remote Indicators - A ground control wire shall be provided for the external indicator circuits listed below:

(1) End of Tape - Indicates warning of end of tape supply. Warning indication shall be 5 minutes tape supply at 3 3/4 ips. Resetting shall not be required nor shall there be any interference with reel installation or removal.

(2) Ready - Equipment is capable of recording within specification.

(3) Recording - Indicates that equipment is recording.

(4) Power ON - Indicates a power on condition.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.4.1.5.4 Internal Controls - The following controls shall be provided within the transport-power supply.

(1) A screwdriver adjustment type of control shall be located in the power supply assembly for control of the regulated +20 VDC supply.

(2) A screwdriver adjustment type of control shall be located in the power supply assembly for control of the regulated -20 VDC supply.

(3) Screwdriver adjustment type of controls shall be located in each channel of the bias adder network subassembly for control of bias and record current.

3.4.1.6 Electrical Connections - Connections to external circuits shall be provided as follows:

| Reference Designation | Receptacle Type | Function |
|---|---|---|
| 1J1 | DPD-W15-34P-1B | Outlet to Remote Control |
| 1J2 | DPD-R15C1-34P-1B | System Interconnection to Unit 2, CH-506/AQH-1(V) |
| 1J3 | DPD-Z14-34P-1B | System Interconnection to Unit 2, CH-506/AQH-1(V) |
| 1J4 | DPD-Z14-34P-1B | System Interconnection to Unit 3, CH-506/AQH-1(V) |
| 1J5 | DPD-R15C1-34P-1B | System Interconnection to Unit 3, CH-506/AQH-1(V) |
| 1J6 | DPD-B8-34P-1B | Primary Power Inlet |

3.4.2 Amplifier Chassis CH-506/AQH-1(V) - The recorder-reproducer amplifier chassis shall meet the following requirements:

3.4.2.1 Function - The amplifier chassis shall have the arrangement and capability of handling seven channels of record-reproduce electronics. Each channel shall accept inputs for either the FM Record-Reproduce Amplifier AM-3770/AQH-1(V), the Direct Record-Reproduce Amplifier AM-3769/AQH-1(V), or the ICS Record-Reproduce Amplifier AM-3773/AQH-1(V). Each channel shall be independent of any other channel and there shall be no interaction between channels.

3.4.2.2 Form Factor - The amplifier chassis shall not exceed the following dimensions:

| Width | Height | Length or Depth |
|---|---|---|
| 8.2 inches | 13 inches | 16 inches |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.4.2.3 Weight - The weight of the amplifier chassis shall not exceed 11 pounds.

3.4.2.4 Contents - The amplifier chassis shall contain the following subassemblies or circuits:

(1) Internal cabling or wiring such that all record-reproduce amplifiers may be interchangeable.

(2) Test plugs shall be provided for testing and troubleshooting purposes.

(3) A separate test jack shall be provided for monitoring the 10 KC frequency reference signal after it has been recorded on the tape. The test jack shall be accessible externally on the amplifier chassis.

3.4.2.5 Controls - There shall be no controls either externally or internally on the amplifier chassis.

3.4.2.6 Electrical Connections - Connections to external circuits shall be provided as follows:

| Reference Designation | Receptacle Type | Function |
|---|---|---|
| 2J1 (or 3J1) | DPD-Z14-34P-1B | Data Signal Input and Output |
| 2J2 (or 3J2) | DPD-R15C1-34P-1B | System Interconnection to MX-4826/AQH-1(V) |
| 2J3 (or 3J3) | DPD-Z14-34P-1B | System Interconnection to MX-4826/AQH-1(V) |

3.4.3 Direct Recorder-Reproducer Amplifier AM-3769/AQH-1(V) The direct amplifier shall meet the following requirements:

3.4.3.1 Concept - The direct record-reproduce amplifier shall be in keeping with a modular design concept. Each direct record-reproduce amplifier shall be identical and electrically and mechanically interchangeable within the amplifier chassis CH-506/AQH-1(V) without the need of readjustments to meet performance requirements stated herein, except for the following normal operating adjustments:



Source: https://assist.dla.mil -- Downloaded: 202640-26T04:33Z
Check the source to verify that this is the current version before use.

(1) Record and reproduce gain adjustments to compensate for variation in head sensitivity

(2) Equalizer adjustments for low, middle and high frequency gain to compensate for variations in head output.

3.4.3.2 Form Factor - The direct record-reproduce amplifier module shall not exceed the following dimensions:

| Width | Height | Length or Depth |
|---|---|---|
| 8.2 inches | 1.7 inches | 12.5 inches |

3.4.3.3 Weight - The weight of the direct amplifier module shall not exceed 1.80 pounds.

3.4.3.4 Contents - The direct record-reproduce amplifier module shall contain a record amplifier subassembly, a reproduce amplifier subassembly and an equalizer subassembly, and shall meet the following detail requirements:

3.4.3.4.1 Input Level - The record amplifier shall be capable of handling an input voltage range of 0.15 Vrms to 3.0 Vrms with the equipment meeting the requirements of paragraphs 3.3.5.3 and 3.3.5.4. The amplifier shall be capable of handling an input of up to 12 Vrms with necessary adjustments.

3.4.3.4.2 Input Impedance - The record amplifier shall provide an input impedance of 40,000 ohms minimum unbalanced to ground with signal ground isolated from chassis ground.

3.4.3.4.3 Output Level - The output level of the reproduce amplifier shall give unity reproduction for input levels to the record amplifier of from 0.15 Vrms to 3 Vrms.

3.4.3.4.4 Output Impedance - The output of the reproduce amplifier shall meet the requirements of this equipment specification when connected to a load impedance of 300 ohms or greater, unbalanced to ground and isolated from chassis ground.

3.4.3.5 Controls - Screwdriver type of controls shall be provided for input and output gain adjustments.



Source: https://assist.dla.mil -- Downloaded: 202610-26T04:33Z
Check the source to verify that this is the current version before use.

3.4.3.6 Electrical Connections - A printed circuit type of connector shall be provided for the record amplifier and a similar printed circuit connector shall be provided for the reproduce amplifier. These connectors shall each mate with a Cinch connector part number 251-15-77-220 or equivalent.

3.4.4 FM Recorder-Reproducer Amplifier AM-3770/AQH-1(V)- The FM amplifier module shall meet the following requirements:

3.4.4.1 Concept - The concept of design and construction shall be in keeping with the modular design concept. All FM record-reproduce modules shall be identical such that they are electrically and mechanically interchangeable within the Amplifier Chassis CH-506/AQH-1(V) in any channel without the need of adjustments to meet the performance requirements of this equipment specification.

3.4.4.2 Form Factor - The FM record-reproduce amplifier module shall not exceed the following dimensions:

| Width | Height | Length or Depth |
|---|---|---|
| 8.2 inches | 1.7 inches | 12.5 inches |

3.4.4.3 Weight - The weight of the FM amplifier module shall not exceed 2.25 pounds.

3.4.4.4 Contents - The FM record-reproduce amplifier module shall contain a record amplifier subassembly, a reproduce amplifier subassembly and an FM filter subassembly, and shall meet the following requirements:

3.4.4.4.1 Input Level - The FM record amplifier shall be capable of meeting the specified performance with an input range of 0.16 to 21.0 Volts rms.

3.4.4.4.2 Input Impedance - The FM record amplifier shall provide an input impedance of 40,000 ohms minimum unbalanced to ground with signal ground isolated from chassis ground.

3.4.4.4.3 Carrier Frequencies - FM carrier frequencies shall be in accordance with IRIG Standards with center frequencies as follows:

| Tape Speed | Center Frequency |
|---|---|
| 30 ips | 54,000 cps |
| 15 ips | 27,000 cps |
| 7 1/2 ips | 13,500 cps |
| 3 3/4 ips | 6,750 cps |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.4.4.4.4 Frequency Modulation Compression - Logarithmic signal compression shall be provided as shown in Figure 1, page 31 wherein an input sine wave signal of .16 volts rms produces a recorded carrier deviation of 6.4 per cent peak and an input sine wave signal of 16 volts rms produces a carrier deviation of 40% peak. Allowable deviation from any point on this curve shall be ± 3 db as measured on the ordinate axis. Zero db corresponds to an input of .16 volts rms and 40 db corresponds to an input of 16 volts rms.

3.4.4.4.5 Compression Constants - After a step change of input signal from .16 V rms to 16 V rms, the recorded rms carrier deviation shall become less than 110 per cent of its final stable value as defined in paragraph 3.4.4.4.4 in not more than two seconds. After a step change of input signal from 16 V rms to .16 V rms, the recorded rms carrier shall become greater than 90 per cent of its final stable value in not more than ten seconds.

3.4.4.4.6 Output Level - The output level from the reproduce amplifier shall be .16 to 1 volt rms into a 200 ohm load for input levels to the record amplifier varying from .16 to 16 V rms. Input levels from 16 to 21 V rms shall produce a 1 V rms output level.

3.4.4.4.7 Output Impedance - The output shall meet the operation requirements of this specification when connected to a load impedance of 200 ohms or greater unbalanced to ground and isolated from chassis ground.

3.4.4.5 Controls - Screwdriver type of controls accessible from the front panel shall be provided for input and output gain adjustments.

3.4.4.6 Electrical Connections - A printed circuit type of connector shall be provided for the record amplifier and a similar printed circuit connector shall be provided for the reproduce amplifier. These connectors shall each mate with a Cinch connector part number 251-15-77-220 or equivalent.

3.4.5 ICS Recorder-Reproducer Amplifier AM-3773/AQH-1(V)- The ICS amplifier module shall meet the following requirements:

3.4.5.1 Concept - The design and construction shall be in keeping with the modular design concept. All ICS record-reproduce modules shall be identical such that they are electrically and mechanically interchangeable within the Amplifier Chassis CH-506/AQH-1(V) in any channel without the need of adjustments to meet performance requirements stated herein except for record and reproduce gain adjustments to compensate for variations in head sensitivity.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.4.5.2 Form Factor - The ICS record-reproduce amplifier module shall not exceed the following dimensions:

| Width | Height | Length or Depth |
|---|---|---|
| 8.2 inches | 1.7 inches | 12.5 inches |

3.4.5.3 Weight - The weight of the ICS amplifier module shall not exceed 2.25 pounds.

3.4.5.4 Contents - The ICS record-reproduce amplifier module shall contain a record amplifier subassembly, a reproduce amplifier subassembly and shall meet the following requirements:

3.4.5.4.1 Input Level - The ICS record amplifier shall be capable of handling an input voltage level of 5 V to 12 V rms.

3.4.5.4.2 Input Impedance - The ICS record amplifier shall provide an input impedance of 40,000 ohms minimum unbalanced to ground and with signal ground isolated from chassis ground.

3.4.5.4.3 Output Level - The output level shall be unity reproduction for an input level to the record amplifier up to 12 V rms.

3.4.5.4.4 Output Impedance - The output of the ICS reproduce amplifier shall meet the operational requirements of this specification when connected to a load impedance of 300 ohms or greater unbalanced to ground and isolated from signal ground.

3.4.5.4.5 ICS Isolation - At the ICS input to the recorder-reproducer there shall be at least 30 db isolation between the ICS signal and the frequency reference signal. At the ICS output from the recorder-reproducer there shall be at least 55 db isolation between the ICS signal and the frequency reference signal. An ICS reference level of 9 V rms shall be used for measuring in both cases.

3.4.5.5 Controls - Screwdriver type controls accessible from the front panel shall be provided for input and output gain adjustments.

3.4.5.6 Electrical Connections - A printed circuit type connector shall be provided for the record amplifier and a similar printed circuit connector shall be provided for the reproduce amplifier. These connectors shall each mate with a Cinch connector part number 251-15-77-220 or equivalent.



Source: https://assist.dla.mil -- Downloaded: 202640-26T04:33Z
Check the source to verify that this is the current version before use.

3.4.6 Mounting MT-3025/AQH-1(V) - This mount shall support the Transport/Power Supply MX-4826/AQH-1(V) and shall meet the following requirements:

3.4.6.1 Form Factor - The transport-power supply mount shall be a center of gravity type and shall not exceed the following dimensions:

| Width | Height | Length or Depth |
|---|---|---|
| 17.16 inches | 15.56 inches | 25.21 inches |

3.4.6.2 Weight - The weight of the transport-power supply mount shall not exceed 11.0 pounds.

3.4.6.3 Vibration Characteristics - The mount shall provide isolation to the transport-power supply when vibrated in accordance with Specification MIL-T-5422.

3.4.6.4 Connector Provision - Provision shall be made on the mount to allow mating connectors for the transport-power supply to be mounted. These connectors shall not be supplied as part of the mount.

3.4.7 Mounting MT-3024/AQH-1(V) - This dual mount shall support and provide vibration isolation for two (2) Amplifier Chassis CH-506/AQH-1(V) and shall meet the following requirements:

3.4.7.1 Form Factor - The dual amplifier chassis mount shall not exceed the following dimensions:

| Width | Height | Length or Depth |
|---|---|---|
| 17.5 inches | 5.2 inches | 17 inches |

3.4.7.2 Weight - The weight of the dual amplifier chassis mount shall not exceed 3.0 pounds.

3.4.7.3 Connector Provision - Provision shall be made to allow the mating connectors for the amplifier chassis to be mounted on the rear of the mount. These connectors shall not be supplied as part of the mounting.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests (3) Acceptance Tests
(2) Production Sample Tests (4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor Demonstration Tests
(2) Service Approval Tests

4.2.1 Contractor Demonstration Tests - Contractor demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.2 Service Approval Tests - At the completion of the contractor demonstration tests and when requested by the procuring activity, the equipment shall be delivered at the contractors expense to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent at the contractors expense to a designated government laboratory for tests. This equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the production



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having test facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests, - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only thoseadjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

| | |
|---|---|
| Temperature | Ambient room |
| Humidity | Ambient room |
| Vibration | Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's |

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following:

| Quantity of Equipments Offered for Acceptance | Quantity to be Selected For Sampling Test |
|---|---|
| First 10 | 1 |
| Next 50 | 1 |
| Next 75 | 1 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.4 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following; except that when a production sample equipment is requested and submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

| Quantity of Equipments Offered for Acceptance | Quantity to be Selected For Life Test |
|---|---|
| First 25 | 1 |
| Next 175 | 1 |
| Next 300 | 1 |
| | 1 for each additional 500 or fraction thereof |



Source: https://assist.dla.mil -- Downloaded: 2026-02-04 07
Check the source to verify that this is the current version before use.

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

| | |
|---|---|
| Temperature | Normal room |
| Altitude | Normal ground (0 - 5000 ft.) |
| Humidity | Room ambient |
| A.C. Voltage (3 Phase) | 115 ± 5 volts (at lowest applicable frequency) |
| D.C. Voltage | 27.0 ± 2.0 volts |

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

## 5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.



Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:33Z
Check the source to verify that this is the current version before use.

## 6. NOTES

6.1 Intended Use - The equipment shall be designed for use in naval aircraft to record and reproduce pertinent data during anti-submarine missions.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltage of 115 ± 1.0 VAC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Bureau of Naval Weapons for consideration. Each request shall be accompanied by complete supporting information.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

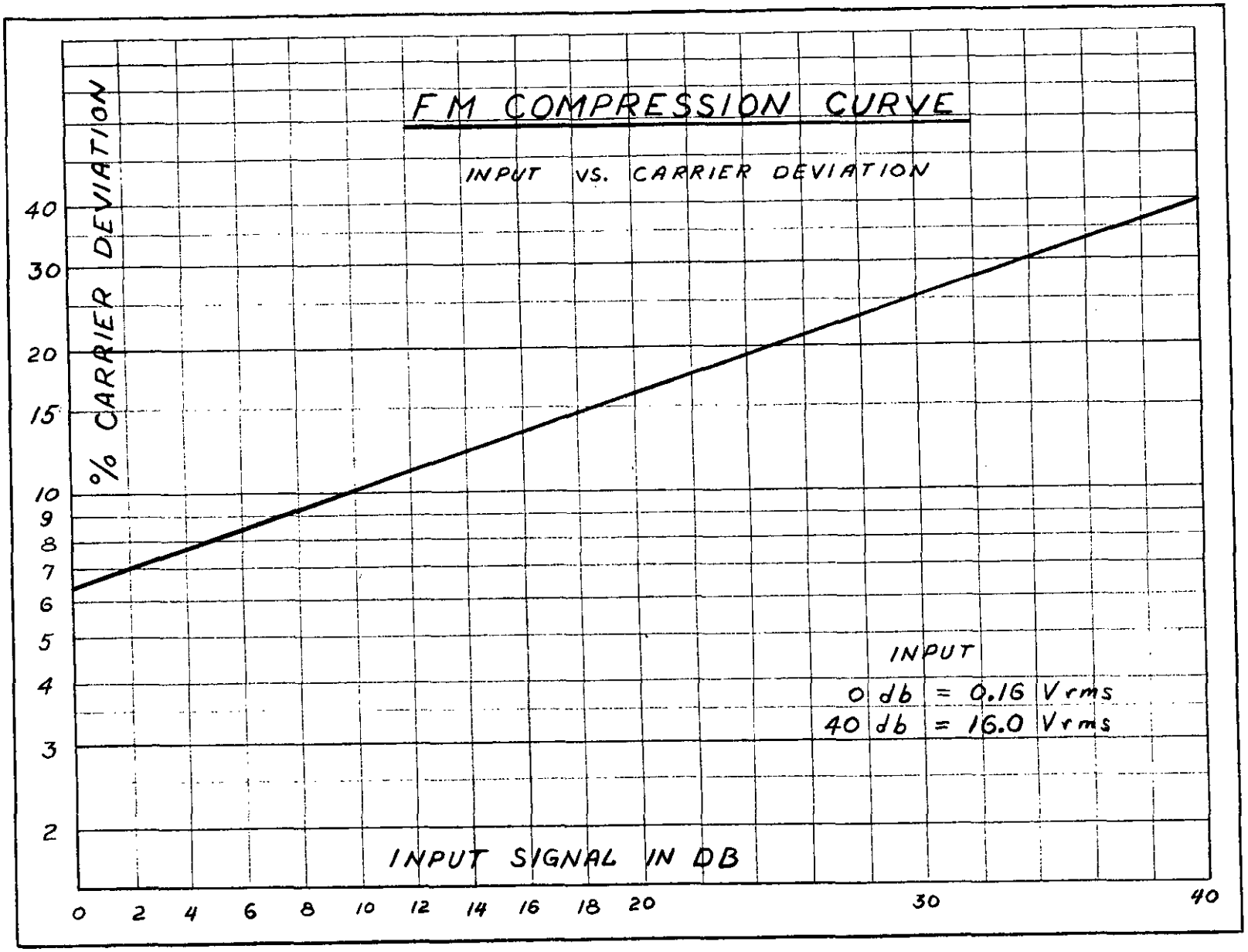



FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

RECORDER-REPORDUCER SET, SOUND AN/AQH-1(V)

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426

Source: https://assist.dla.mil -- Downloaded: 20260-26104:33Z
Check the source to verify that this is the current version before use.

C-8278